

From Abstracts to Rooms: Poster Session Assignment by Semantic Similarity and Optimisation



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Declaration

I declare that the work presented in this dissertation is, to the best of my knowledge and belief, original and my own work. The material has not been submitted, either in whole or in part, for a degree at this, or any other university. Estimated word count is: **13234**

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A dissertation submitted for the degree of *Master of Science* in Cyber Security.
August, 2026

Abstract

This is the beginning of the abstract. According to the Postgraduate Research Regulations 2022-23, it should not be any longer than 300 words. However, as your dissertation is to be submitted as part of a taught Master programme (which includes the MRes), there is no strict limit imposed on the abstract length. For general guidance on language, presentation, and ordering of your dissertation, refer to Appendix 2 (Pages 30-33) <https://bit.ly/2Q4H43I>.

Acknowledgements

Acknowledgements you may want to make. (this is not required when submitting your dissertation before your viva and you can add a dedication in your final dissertation after your viva if you wish.)

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Chapter 1

Introduction

1.1 Motivation

1.2 Problem Statement

1.3 Relevance

1.4 Proposed Approach

1.5 Contributions

1.6 Summary of Results

1.7 Did It Work?

1.8 Dissertation Structure

Chapter 2

Literature Review

2.1 Conference Session and Poster Assignment

The problem of organizing a scientific conference by computational means is not new. In an early study, Eglese and Rand (1987) surveyed the attendees of a 1985 conference to express which seminars they wished to attend, and then built a schedule that would allow the fulfillment of as many of these wants as possible. From that point on, the organizing of poster placements in a conference has become received a great deal of attention leading to the establishment of a large amount of operational research literature, where this problem has mainly been dealt as an integer linear problem. Research such as Stidsen, Pisinger, and Vigo (2018) describe a tool used to schedule the EURO-k conferences, events with more than two thousand presentations, by using optimization models to place related research buildings close to each other. Riquelme et al. (2022) focused on a "track-based" version of the problem—sorting choices into specific paths. Instead of finding an exact optimal solution, which is always computationally slow, they used a "simulated annealing heuristic," which is a smart shortcut algorithm. This shortcut found comparable solutions in a fraction of the time. Closest to our setting, Bulhões, Correia, and Subramanian (2022) define a general session scheduling problem whose objective is to maximize the similarity of papers put next to each other. They prove that this problem is NP-hard, and solve real conference instances with branch-and-cut and branch-cut-and-price methods. While this exact approach guarantees an optimal solution, it is computationally expensive, leaving room to develop faster methods that produce comparable results.

There is also a smaller line of research that finds topics by reading the actual paper texts, not by relying on pre-set labels or user preferences. Burke and Sabatta (2015) use Latent Dirichlet Allocation (LDA) directly to the submitted documents to learn the conference topics. They then schedule the parallel tracks so that simultaneous sessions cover the most different topics possible. By maximizing the distance between session topics, they ensure attendees are not forced to choose between two similar talks. In current practice, large

venues face the same task at much greater scale: For example, CVPR (CVPR, 2024) groups its accepted posters by similarity within each session. To decide which posters are similar, they use SPECTER embeddings and then reduce the size of those vectors down to a single line with one-dimensional t-SNE. Posters that land close together on that line are put in the same group. This is, to our knowledge, the closest real-world precedent for the problem we study; however, it is a projection heuristic rather than an explicit optimization, and the quality of the resulting layout is not measured.

Put together, the literature has largely pursued conflict avoidance and preference satisfaction. The topical coherence of a capacity-limited physical room, treated as an explicit optimization objective in itself, remains largely unexplored.

2.2 Representing Scientific Text as Vectors

Sentence embedding models map a text to a fixed-length vector such that semantically similar texts receive nearby vectors. Sentence-BERT introduced the now-standard approach of fine-tuning a pretrained transformer with siamese network structures, producing sentence vectors that can be compared directly with cosine similarity (Reimers and Gurevych, 2019). The model used later in this work, all-MiniLM-L6-v2, combines this training scheme with the MiniLM distillation method, which compresses a large teacher transformer into a six-layer student with 384-dimensional outputs (Wang et al., 2020). In contrast, a family of models exists specifically for scientific documents: SPECTER fine-tunes SciBERT on the citation graph, treating papers that cite one another as similar (Cohan et al., 2020); its successor SPECTER2 is trained on over six million citation triplets and attaches task-specific adapters (Singh et al., 2023); and Aspire models fine-grained similarity by matching individual sentences between papers (Mysore, Cohan, and Hope, 2022). The landscape is wider than the three models examined later: contrastive objectives such as SimCSE learn general-purpose sentence embeddings without any labelled pairs (**gao2021simcse**), and SciNCL replaces SPECTER’s discrete citation relations with sampling over citation-graph embeddings (**ostendorff2022neighborhood**). These models were trained on different signals and therefore encode different notions of what makes two papers similar, a distinction whose practical consequences we examine in Chapter 5.

2.3 Hierarchical Clustering of Documents

Clustering methods vary not solely in quality but in their structural properties, and these properties determine how well a method fits a given task. Hierarchical agglomerative clustering starts from every point as its own group and repeatedly merges the closest pair, producing a tree that can afterwards be cut at any level to yield any number of clusters; Ward’s method chooses each merge so that the increase of the within-group variance is

minimal, which favours compact groups (Ward, 1963). The k-means algorithm instead partitions the data into a number of groups that must be fixed in advance, and because it starts from a random initialization, its result can vary from run to run, which reduces reproducibility (MacQueen, 1967; Jain, 2010). Density-based methods such as HDBSCAN extract clusters of varying density from a hierarchy of density estimates (Campello, Moulavi, and Sander, 2013; McInnes, Healy, and Astels, 2017); a defining feature of this family is that points in low-density regions are labelled as noise and left outside every cluster. Finally, probabilistic topic models such as Latent Dirichlet Allocation (LDA) describe each document as a soft mixture of topics rather than assigning it to exactly one group (Blei, Ng, and Jordan, 2003).

In this work, clustering is judged by practical fit rather than benchmark accuracy: every poster must be placed (no unassigned outliers), the number of groups must be freely re-cut to match physical room capacities, and the outcome must be deterministic so that our layout heuristics remain stable. Chapter 6 derives the choice of hierarchical clustering directly from these three requirements.

2.4 Capacitated Grouping and Clique Partitioning

At its combinatorial core, our task is the clique partitioning problem: assign items to mutually exclusive groups so that the total similarity within groups is maximized. (Grötschel and Wakabayashi, 1989) introduced this formulation as a clustering model and proposed cutting plane methods for its solution. The problem was later shown to be NP-hard (Grötschel and Wakabayashi, 1990), and the geometric structure of its feasible solutions has been studied extensively (Oosten, Rutten, and Spieksma, 2001). Even with modern commercial solvers, an exact solution becomes impractical as instances grow, something that (Du et al., 2022) has also concluded through a direct comparison of models and solvers on clique partitioning benchmarks. Our problem in Chapter 7 adds two further restrictions to this core, namely room capacities and a minimum occupancy per room, and regardless of the added rules, it still doesn't remove its NP-hardness.

2.5 Metaheuristics for Partitioning Problems

For partitioning problems, a widely accepted set of metaheuristics exists. GRASP repeatedly builds a solution with a randomized greedy construction and then improves it with local search, keeping the best result over many restarts (Feo and Resende, 1995). Tabu search follows a single trajectory and uses a short-term memory of recently made moves to escape local optima (Glover and Laguna, 1997). Variable neighbourhood search alternates between descending to a local optimum and perturbing the solution with systematically growing shakes (Mladenović and Hansen, 1997). Memetic algorithms evolve a population of solutions

through recombination and mutation, with local search applied to the offspring (Moscato and Cotta, 2003). These methods have been applied to clique partitioning itself, for example through iterated tabu search (Palubeckis, Ostreika, and Tomkevičius, 2014), and hybrid approaches additionally exist that embed an exact solver inside a heuristic loop, a design we will go back to in Chapter 7. The metaheuristic family is of course wider than these five; simulated annealing, for instance, has been applied in the conference scheduling setting itself as well (Riquelme et al., 2022).

2.6 Benchmarking Algorithms

All of the methods above are randomized, so a single run of each says little about which one is better. The established standard procedure is therefore to compare distributions of results over multiple independent runs, and to test the differences with non-parametric statistics, since the outcomes of heuristics cannot be assumed to follow a normal distribution. Demšar (2006) recommends exactly this design: the Wilcoxon signed-rank test for comparing two methods on paired results, and the Friedman test as an omnibus test when several methods are compared at once. Running many tests together creates a problem of its own: each test has a small chance of a false alarm, so across ten pairwise comparisons at least one “significant” result is quite likely to be fake. The procedure of Holm (1979) fixes this by making each individual test stricter, keeping the whole table at the usual five percent error level, and García and Herrera (2008) show how to apply it when every method is compared against every other, which is our case. Chapter 7 follows exactly this protocol; it is also why every method there receives the same time budget and the same random seeds, since the tests compare methods seed by seed, and that pairing is only fair if both methods faced identical conditions on each seed.

2.7 Summary and Positioning

Together, the literature is divided into three paths. First, the operational research norm schedules conferences well, but its objectives are preventing overlapping sessions and satisfying preferences, not the topical coherence of a room. Second, a smaller line of research reads the papers themselves to find topics but stops at abstract groupings without rooms or seat capacities. Third, in practice, large venues already assign posters with embedding-based heuristics, without measuring the quality of the result. All building blocks our pipeline needs exists on its own: sentence embeddings, hierarchical clustering, the clique partitioning problem, metaheuristics, and a statistical protocol for comparing them fairly. What no prior work does, is combine them, a reproducible pipeline that takes a conference’s raw submissions and returns a topically coherent, physically feasible assignment of papers to capacity-limited rooms. That arrangement is what this dissertation provides.

Chapter 3

Pipeline Overview

3.1 Design Goals

The pipeline was built to meet five requirements. First, the layout it produces must be topically coherent: papers found in the same room should talk about similar concepts, since this is the entire purpose of the system itself. Second, the layout must be physically feasible, meaning that no room is allowed to hold more papers than it has seats and no room is allowed to be left empty. Third, the pipeline must maintain peak performance at the scale of a real conference, which today means thousands of papers. Fourth, it must be general: the same pipeline, standardized, should accept any conference’s papers and any set of room sizes, ensuring reliability at any scale of a scientific conference. Finally, it must be reproducible: every random choice the pipeline makes is controlled by a fixed seed, so repeated runs follow exactly the same search. However, in the heuristic faze only the search is stopped by wall-clock time rather than by a step count, the resulting layout is the same in practice but not guaranteed to be identical, a point Chapter 8 returns to.

3.2 Architecture

The pipeline is made up of four stages, shown in Figure 3.1. First stage obtains the accepted papers of a conference, in our case the abstracts and corresponding metadata of the accepted NeurIPS 2025 submissions. Embedding starts the second stage by converting each abstract into a fixed-length numerical vector, so that the similarity of two papers can be computed as the similarity of their vectors. Clustering groups these vectors into many small topic clusters, by calculating the distance of the vectors with eachother and grouping the closest ones together. Room assignment then packs whole clusters into the available rooms, maximizing the similarity of the clusters that end up together, and an evaluation step measures the quality of the finished layout. Each stage hands its output to the next: stage one delivers the

abstracts, stage two turns them into unit-length vectors, stage three groups the vectors into clusters with centroids, and stage four turns the clusters into a paper-to-room layout again based on similarity of papers between clusters.

The four stages follow the standard data science pipeline of collection, representation, modelling and validation. As that framework itself emphasizes, the process is iterative rather than linear: in our pipeline, if clustering produces a cluster larger than the largest room, no feasible layout can exist, and the flow loops back to re-cut the same cluster tree more finely until every cluster fits. The pipeline is demonstrated throughout this dissertation on a single real instance, the 5,286 accepted papers of NeurIPS 2025 assigned to twelve rooms, and the properties of this dataset are described in Chapter 4.

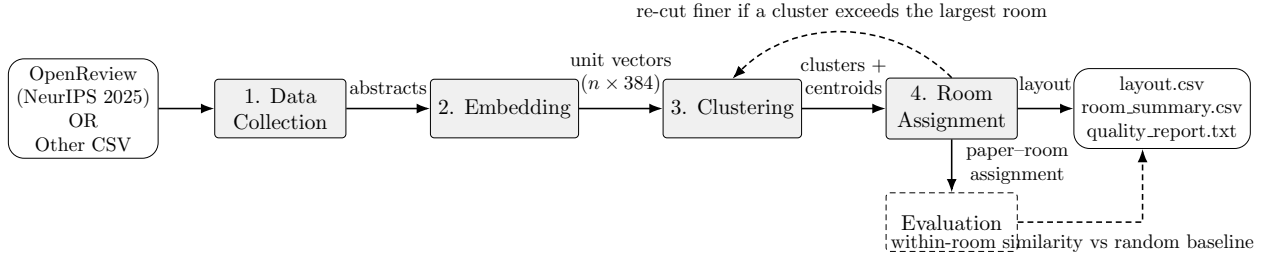


Figure 3.1: The four stages of the pipeline. The input is any conference’s papers as a csv file with one abstract per paper. NeurIPS 2025, retrieved from OpenReview, is the dataset used throughout this dissertation. The dashed arrows show the two non-linear parts: the evaluation of the finished layout against a random baseline, and the loop that re-cuts the cluster tree when a cluster does not fit the largest room.

3.3 Why This Decomposition

The central design decision of the pipeline is that papers are not assigned to rooms individually. Instead, they are first grouped into clusters, and the optimization then places the clusters into the given rooms. Three reasons for doing this. The first is scale: assigning 5,286 individual papers to twelve rooms encompasses a search space far greater than what any exact or heuristic method can explore meaningfully in a short time, while a few hundred clusters can simplify the task significantly. The second is signal: a single paper provides little topical information on its own, whereas a cluster of related papers can end up having a stable, recognizable subject, making it easier to optimize. The third is past experiences: nobody walks a poster session with a list of single papers in mind. People think in topics, see the robotics posters, then the medical ones, so the natural focus of a session area the paper subject areas, and the produced clusters end up being building blocks of those areas. Assigning papers one by one could sometimes scatter a topic across several rooms, assigning

whole clusters increases the likelihood of every topic being in the right place.

This decomposition has an honest cost, and we state it here because it is a property of the shape rather than of any individual method: clusters are unchanging. Once formed, a cluster is decided, so the grouping might interfere with the flexibility of splitting a topic across two different rooms. This lack of flexibility is what later makes filling every room to its exact seat count difficult, and its consequences are examined in Chapter 9. Further details on which clustering method forms the clusters, and which optimization method packs them, are also treated in Chapter 6 and Chapter 7 respectively.

3.4 Notation and Terminology

The following notation is used throughout. The conference has n papers and R rooms with seat capacities $c_1, \dots, c_R$. Clustering produces C clusters with sizes $s_1, \dots, s_C$, each represented by a centroid, and W denotes the $C \times C$ table of pairwise centroid similarities. A layout is an assignment that maps every cluster to exactly one room. Two terms are kept strictly apart: the *objective* is the cluster-level score that the optimization maximizes, while the *within-room similarity* is the paper-level measure by which the finished layout is assessed. These are all related but not identical measurements, and further details on the importance of each is analysed in Chapter 9. However, the formal optimization model built from this notation is stated at the start of Chapter 7.

3.5 Evaluation Strategy

The pipeline is evaluated using three distinct measures. The first is the optimization objective, which serves as the internal mathematical score that the assignment methods in Chapter 7 compete to maximize. The second is the mean within-room similarity of the actual papers. Because this is calculated independently of the optimizer, it acts as an unbiased reality check of the grouping quality. The third is a baseline control, created by randomly shuffling the exact same papers into the same room sizes. Because all papers at a single conference naturally share a broad topic and vocabulary, any random grouping will inherently yield a baseline level of similarity. Consequently, a raw similarity score is meaningless on its own; it only becomes interpretable when compared directly against this random baseline to prove the algorithm is genuinely finding deeper connections.

Chapter 4

Data Collection

4.1 Two Datasets and Their Roles

This work uses two datasets with strictly separated roles; Table 4.1 compares them. Every experiment and every design decision, from the model comparison of Chapter 5 to the final layout of Chapter 8, is made on the NeurIPS 2025 corpus. The arXiv sample is used exactly once, in Chapter 8, to run the finished tool on a corpus it was not developed on; nothing touches it before that run. A third, small set is derived from the NeurIPS data: the paraphrase set of Section 4.3.

Table 4.1: The two datasets side by side.

	NeurIPS 2025	arXiv sample
Source	OpenReview API	arXiv metadata snapshot
Papers	5,286	10,000
Fields	5	4
Missing values	none	none
Fields the tool reads	Title, Abstract	title, abstract
Reference labels	Primary Area	group, subcategory
Abstract mean	181.6 words	138.4 words
Abstract range	37–516 words	5–344 words
Categories	13 primary areas	9 arXiv groups
Balance	heavily unbalanced	close to balanced
Snapshot	July 2026, frozen	frozen sample, fixed seed
Role	all development	demonstration only

4.2 The NeurIPS 2025 Dataset

The papers come from OpenReview, the platform on which NeurIPS runs its review process. A short Python script, using the official OpenReview client library, requests all submissions carrying the venue identifier `NeurIPS.cc/2025/Conference` and writes them to one CSV file, one row per paper. The accepted NeurIPS 2025 submissions were downloaded in July 2026, and the dataset reflects the state of the platform at that time. Only the abstract feeds the pipeline; the title labels the outputs, and the rest is kept as reference only.

Figure 4.1 shows the categories the authors chose for their papers. Deep learning alone holds 1,685 papers, almost a third of the conference, while the three smallest areas hold fewer than sixty each. Because most papers here are already similar, our implementation can show false success thus leading to the need of a second testing dataset.

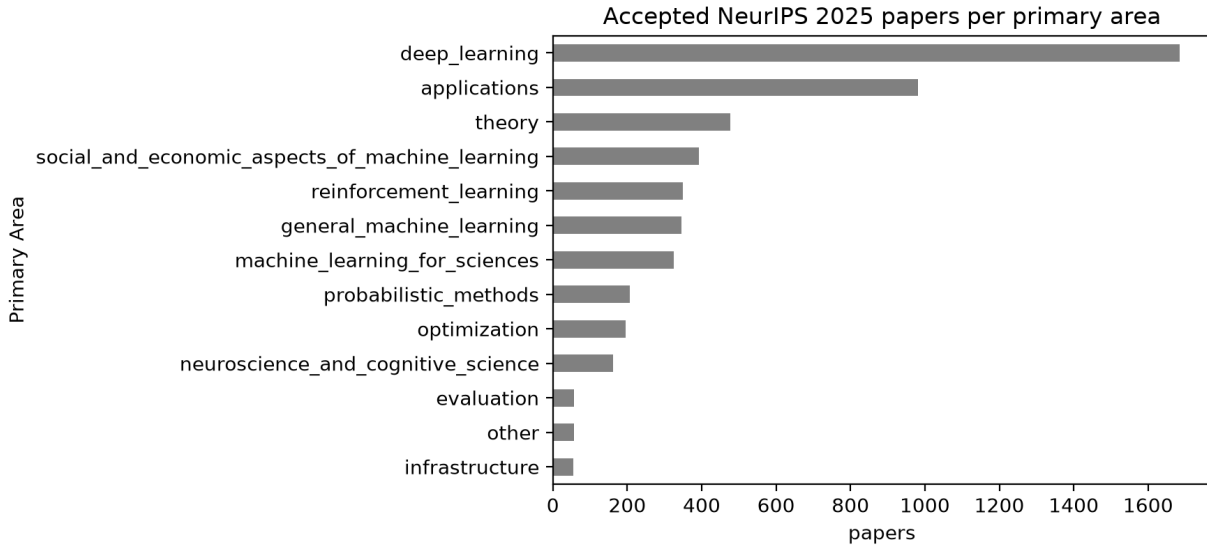


Figure 4.1: Accepted NeurIPS 2025 papers per primary area, from 1,685 papers down to 54.

4.3 The Paraphrase Evaluation Set

Chapter 5 needs a labelled test set for its model comparison. We built one from the NeurIPS data: 250 random abstracts, each rewritten into a single sentence by the Gemma E2B model, run locally through Ollama. A good model should recognise the original paper from that sentence. The words change, the meaning does not. Rerunning the model gives different sentences each time, so the set was written once and kept unchanged.

4.4 The Demonstration Dataset

The NeurIPS corpus is topically narrow, almost a third of it is deep learning. High within-room similarity could come partly from the papers resembling each other, not from the implementation being good. The finished application is therefore also run once, deliberately, through different dataset, built as Table 4.2 describes. Figure 4.2 shows the outcome: nine fields, close to balanced, the opposite of Figure 4.1.

Table 4.2: Construction of the demonstration dataset.

Source	arXiv metadata snapshot, ~3 million entries
Selected	the nine top-level arXiv groups
Category used	primary, the first one the authors listed
Excluded	legacy categories from the 1990s
Cap per group	11,000, the size of the smallest group
Within a group	cap split equally among subcategories; undersized ones contribute all they have
Parent set	[TO FILL] papers
Sample	10,000, uniform random, fixed seed
Provided	sampled file ships with the code

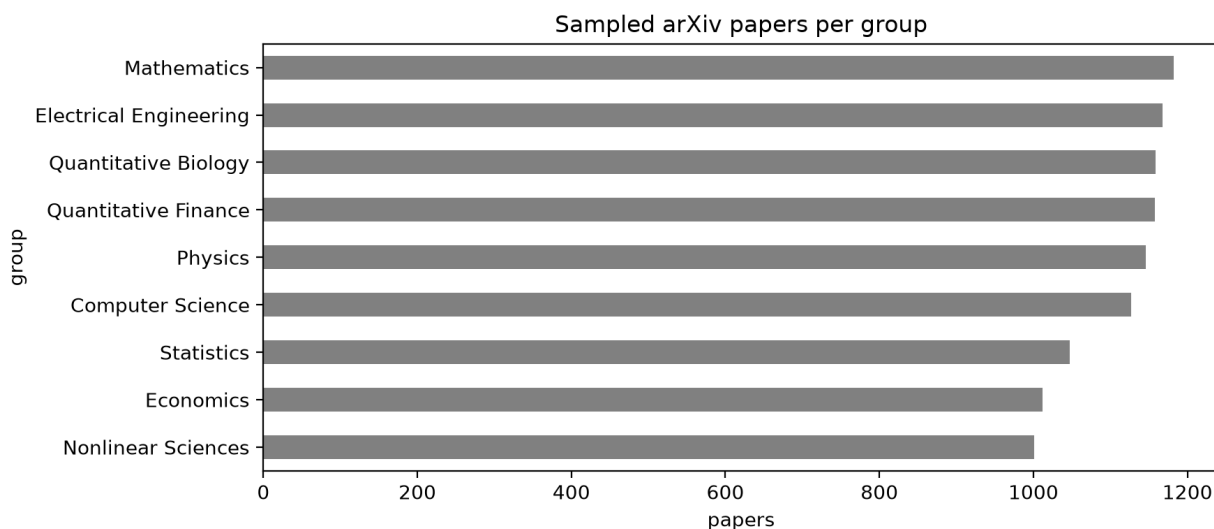


Figure 4.2: The 10,000 sampled arXiv papers per group, from 1,182 down to 1,001.

Chapter 5

Embedding Model Selection

5.1 The Candidates

The pipeline turns every abstract into a vector, a task assigned to the final chosen embedding model. This chapter describes how that choice was made. An embedding model reads a sentence and looks at how the words work together to figure out its exact topic. Based on that topic, it outputs a numerical value called a vector. It does this for every piece of text you input, and in the end, the vectors that are closest to each other represent the most similar topics. The abstract is the text given to it, because the abstract is the summary the authors wrote themselves and therefore carries more information about the topic per word than any other part of the paper. Additionally, and because it is short, under two hundred words on average against several thousand for a full paper, it keeps the work small enough for the pipeline to compute the layout in better time and typically without sacrificing quality.

This study compares three candidates, which are listed in Table 5.1. Two of them were built specifically for scientific documents. SPECTER2 learns from citation signals, meaning it treats papers that cite one another as similar (Cohan et al., 2020; Singh et al., 2023), while Aspire learns from the sentences in which two papers are cited together (Mysore, Cohan, and Hope, 2022). The third candidate, all-MiniLM-L6-v2, is a small general-purpose model from the Sentence-BERT family (Reimers and Gurevych, 2019; Wang et al., 2020). It was trained on scientific text as well, but on everyday text alongside it (Sentence-Transformers, 2021), which is expected to help wherever papers describe their work in ordinary rather than technical language, as those in economics and in the social and economic areas of machine learning often do. This setup is built on purpose to test two specialized models against one single model. While the two specialists are each trained on only one kind of language, the single model is trained on both at the same time, allowing us to see which approach works best.

Table 5.1: The three candidate models, each used exactly as delivered.

Model	Size	Dim.	Window	Metric
all-MiniLM-L6-v2	22.7M params	384	256 tokens	cosine
SPECTER2	BERT-base	768	512 tokens	Euclidean
aspire-sentence-embedder	BERT-base	768	512 tokens	Euclidean

5.2 Why These Three

A large variety of models can turn text into vectors, but only a filtered amount is right for our task. Three questions do the narrowing: which models can perform this task at all, which types of those models are best, and what the available hardware can run.

The first question removes most models. A language model such as BERT is trained by hiding words in a sentence and having the model guess them. This teaches it a great deal about language, but it never teaches it to arrange texts according to how similar they are, and Reimers and Gurevych (2019) showed the consequence: comparing the output of such a model directly gives poor results, worse even than far simpler methods that ignore word order altogether. A model becomes usable here only after a second stage of training, in which it is shown pairs of related texts and taught to keep them close together while pushing unrelated texts apart. This is why SciBERT, the standard model for scientific text (Beltagy, Lo, and Cohan, 2019), was not considered. Reading science is not the missing part: SciBERT simply never received that second stage, and comparing texts is the only thing it would be needed for.

The second question is which type of models that can do the task do we actually select as best. Among the models that did receive that training, some were trained on scientific papers in particular, and were taught which papers belong together by looking at how researchers cite one another. Models built for text in general are trained on a broad mixture of everyday writing, in which scientific papers and citation data appear only as a small part rather than as the whole. The literature reports that the scientific models do better on scientific tasks for this reason (Singh et al., 2023), and the papers used here are scientific abstracts, so they ought to have the advantage. Whether they actually do is what the accomplished experiment tried to establish, and answering a question of that form does not require many models, only a strong representative of each kind.

The three candidates follow from that. all-MiniLM-L6-v2 represents the models built for text in general. It is the standard small model of the Sentence-BERT family (Reimers and Gurevych, 2019; Wang et al., 2020) and was downloaded more than 160 million times to the date of this study being written (Sentence-Transformers, 2021). It is general only in the sense of its intended use, however, and not in the sense of never having read science: its training mixture contains a substantial body of scientific papers, along with a portion of the very citation data that SPECTER was built on (Sentence-Transformers, 2021). This turns

out to matter for the result, and is returned to in Section 5.5. It is also the smallest model in the comparison, which makes it a challenging competitor when it comes to compute time. SPECTER2 represents the scientific models: it is the successor of SPECTER, the best known model of its kind (Cohan et al., 2020), begins from SciBERT, and is then trained on more than six million pairs of papers in which one cites the other, with its own documentation recommending it over the original for new work (Singh et al., 2023). Aspire is a second scientific model, included because it learns from a different signal, namely the sentences in which two papers are mentioned together, which state in words how the two are related (Mysore, Cohan, and Hope, 2022). If only one scientific model had been tested, a poor result could always be blamed on that one model. Testing two that learned in different ways means the outcome says something about scientific models in general, and not about one of them in particular.

The third question is practical. Every candidate had to be free to use and had to run on the available laptop graphics card, which has 6 GB of memory. This excludes the commercial embedding services, which charge for use, limit how much text may be sent, and can be replaced by their provider at any time, so that the same experiment run later might no longer give the same result. It also excludes models too large to fit on the card.

5.3 The Test

The test uses the paraphrase set described in Section 4.3, which contains 250 abstracts that were each rewritten as a single sentence in plain everyday English. The idea behind it is simple. A model is shown one of these sentences and asked to find the paper it was written from, among all 5,286 papers in the dataset. If the model understands what the sentence means, the correct paper should appear at or near the top of its list, so the position at which that paper appears is recorded and called its rank. A rank of one means the model found the right paper immediately, and lower ranks are therefore better.

In practice, all 5,286 abstracts are embedded once and the 250 rewritten sentences are embedded once, after which each sentence is compared against every abstract and the abstracts are sorted by how close they are to it. The 250 papers are split into five experiments of fifty, with no paper used twice, so that it becomes visible whether a model is consistently better or merely fortunate on one particular set. Each model is then described by three numbers: the average rank it achieves, the number of times it reaches rank one, and the time it needs to embed the dataset.

For a comparison to be meaningful, each model must be given the same job, otherwise a model might win simply because it was given an easier one. Every model therefore received exactly the same text: the abstract, on its own, with the title left out. SPECTER2’s documentation suggests giving it the title together with the abstract, but doing so would have given one model more of the paper than the others, and any difference in the results could then no longer be attributed to the model itself. Every model was also timed on the

same graphics card, and the scripts stop with an error rather than moving the work to the processor if that card is unavailable, since a model timed on different hardware could not be compared with the others at all.

Each model was, however, allowed to compare texts in its own way. Every model measures how close two pieces of text are, but not all of them measure it in the same manner: MiniLM compares the direction of two vectors, whereas SPECTER2 and Aspire compare the distance between them, which is how each was trained. Each model was therefore judged by the measure its own authors intended, since a model asked to use a different one would be at a disadvantage that has nothing to do with its understanding of the text.

5.4 Hardening the Test

The first version of this test did not work, and the failure is described here because repairing it forms part of the method. In that version the paraphrases were written at the same length and in the same style as the originals, and all three models placed every one of the 250 papers at rank one. A test that everything passes is of no use, since telling the models apart is the only thing it has to do.

The primary issue was the accessibility of key scientific words. A search using the distinctive technical terms of an abstract still returned the correct paper at rank one, while a search using only plain everyday language did not. The models were therefore matching words rather than meaning, which is far easier than it appears: an unusual technical term occurs in only a handful of abstracts out of 5,286, so a paraphrase that keeps such terms can be traced back to its original without understanding anything about it. The instruction given to the paraphrase model was rewritten so that its wording could not be reused:

`Rewrite the following scientific abstract as ONE single sentence using only plain everyday English. Imagine you are explaining this paper to a smart 14-year-old who has never read a scientific paper.`

`Strict rules: do not reuse any technical terms, jargon, method names, model names, algorithm names, dataset names, acronyms or proper nouns from the original; replace any specialized vocabulary with plain descriptions; use common everyday words throughout; write exactly ONE sentence; capture what the paper does, why it matters, and what it found; return only the rewritten sentence.`

A small trial of 25 paraphrases confirmed that the models no longer scored perfectly and could be told apart. Only then was the full set of 250 generated, frozen, and reused unchanged, as described in Section 4.3.

5.5 Results

Table 5.2 reports the final results. MiniLM achieves the best average rank, 96.64 against 193.81 for SPECTER2 and 221.16 for Aspire, and it does so in each of the five experiments separately rather than only on average. It also finds the correct paper immediately more often than either specialist, 76 times out of 250 against 48 and 44, and it embeds the whole dataset in 7.1 seconds where the specialists need over twenty minutes, a difference of more than a hundred and fifty times. No statistical significance test was run on this comparison, because the differences are large, point in the same direction in every one of the five experiments, and are accompanied by a difference in running time that would decide the matter on its own.

Table 5.2: Results over the 250 paraphrases. Lower rank is better.

Model	Avg. rank	Rank-1 hits	Embed time
all-MiniLM-L6-v2	96.64	76/250 (30.4%)	7.1 s
SPECTER2	193.81	48/250 (19.2%)	1,236.2 s
aspire-sentence-embedder	221.16	44/250 (17.6%)	1,126.4 s

That a general-purpose model should beat two specialists is a surprising result, and its explanation lies in what the model was trained on. MiniLM is not purely general: of the 1.17 billion pairs of texts it was trained on, roughly 210 million, about eighteen percent, come from a corpus of scientific papers, together with a set of citation pairs from SPECTER itself (Sentence-Transformers, 2021). It has therefore read a great deal of science while also reading a great deal of ordinary English. The authors of SciRepEval report the same pattern from the other direction, finding that general-purpose models outperform SciBERT on their benchmark and attributing this to exactly this kind of scientific data appearing in general training sets (Singh et al., 2023). SPECTER2 and Aspire, in contrast, learned only from scientific citation signals (Singh et al., 2023; Mysore, Cohan, and Hope, 2022). The queries used in this test are written in plain English while the papers they must be matched to are written in technical language, which suggests that a model familiar with both kinds of text is better placed to connect them. The test differs in this respect from the task the tool actually performs, where abstracts are compared with other abstracts rather than with plain sentences, and this difference is taken up in Chapter 10.

5.6 The Decision

The model of choice is therefore all-MiniLM-L6-v2, and it meets both of the criteria set out at the beginning of this chapter. It is the more accurate of the three, achieving the best average rank overall and in each of the five experiments separately, so accuracy is not being given up in exchange for speed. It is also the faster model by a margin that matters in

practice rather than merely on paper. The tool embeds every collection of papers it is given, and the difference between seven seconds and twenty minutes is the difference between a tool an organiser can run again and again while trying out different room sizes, and one that has to be started and returned to later.

Each model was used exactly as its authors released it, including the amount of text it reads at once, which is 256 word pieces for MiniLM and 512 for the other two. MiniLM therefore reads slightly less of the longest abstracts than the other two models do, and it produced the better result nonetheless.

Chapter 6

Clustering

6.1 What the Grouping Must Do

Chapter 5 finished with every paper turned into a vector, and Chapter 7 will begin with small groups of papers being packed into rooms. This chapter is the intermediate stage between the two: it takes the 5,286 papers and sorts them into small groups of similar papers, and these groups become the building blocks of the layout. What similar means was set forth in the previous chapter: two papers about the same topic receive vectors that lie close together, so grouping papers by topic is now a question of which vectors that are close to each other in value.

The reason for grouping at all is the size of the problem. Suppose the papers were placed one at a time. Each paper can go into any of the twelve rooms, so there are thousands of choices to make, and the choices depend on each other: every paper that takes a seat leaves one seat fewer for all the papers still waiting, and a room filled early with the wrong papers cannot be undone without moving everything placed after it. Finding the best layout would mean trying combinations of these choices, and the number of possible combinations is beyond what any computer could search through. Grouping first cuts the problem down to a manageable size: instead of asking where each of 5,286 papers should go, the next chapter only has to ask where each of a few hundred groups should go, and a question of that size can be answered well.

This type of approach is an established idea already. In route planning, a well known family of methods first groups the customers by location and then plans a route within each group, specifically because the full problem is too large to solve in one piece, and this approach produced excellent results and was extended by many later works (Fisher and Jaikumar, 1981). The same two-step idea has also been applied to conference scheduling itself (Bulhões, Correia, and Subramanian, 2022).

The grouping must fulfill three requirements. Every paper must end up in a group, because every paper needs a seat. Every paper must end up in exactly one group, because a

paper can only stand in one room. And the method must work unchanged on any conference the tool is given, not only on this one.

6.2 The Method: Ward's Hierarchical Clustering

The method chosen is hierarchical clustering with Ward's rule (Ward, 1963). The idea is a loop that does one thing over and over. At the start, every paper stands alone, so there are 5,286 groups of one. At every step, the method looks at every possible pair of groups and asks the same question for each: if these two were merged, would the papers of the new group still sit close around their average, or would they be spread out? It then performs the one merge where the papers stay closest together, writes that merge down, and repeats. Ward's rule is the name of this test (Ward, 1963): it always tries the merge first in imagination, measures how spread out the combined group would be, and only carries out the merge that adds the least spread. After 5,285 such merges everything has been joined into one, and the written record of all the merges forms a tree (Murtagh and Contreras, 2012).

The number of groups follows one fixed rule, applied unchanged to any dataset the tool is given: the number of papers divided by ten, rounded to a whole number. For this conference that gives $5,286 / 10 = 529$ groups. Why ten is the right divisor is explained in the next section.

Three properties of this method fit the task exactly. The first is that the tree is built once and can then be cut at any group count without building anything again. Table 6.1 shows what this costs in practice: building the tree for all 5,286 papers takes about two seconds, and cutting it takes six thousandths of a second, so trying a different number of groups is essentially free. The implementation saves the tree to disk and re-cuts it instantly. The second is that the method contains no randomness: the same papers produce the same tree and the same groups every single time it is run. This matters beyond convenience, because it makes every number in this dissertation reproducible, which is one of the goals set in Chapter 3. The third is that it adjusts to the data on its own: the group count is computed from the number of papers rather than fixed by hand, and since cutting is free, the very same tree serves whatever count a corpus requires.

Table 6.1: What the clustering costs, for all 5,286 papers.

Step	Time
Building the tree	1.99 s
Cutting it into 529 groups	0.006 s
Full script, everything included	3.57 s

Three common alternatives were considered and set aside, and Table 6.2 lists what each one is missing. This is not a claim that these methods are worse in general; each simply

lacks one thing this task cannot give up. K-means is the most widely used clustering method (MacQueen, 1967; Jain, 2010). It must be told the number of groups before it starts, and it begins from starting positions chosen at random, so running it twice on the same papers can produce two different groupings. In contrast, Ward needs no number in advance and contains no random step, so its result is the same on every run. HDBSCAN removes the need to choose a number of groups, as it finds them on its own (Campello, Moulavi, and Sander, 2013; McInnes, Healy, and Astels, 2017). It does this, however, by declaring papers that fit nowhere to be noise and leaving them out, and a poster session cannot do that: every paper must receive a seat. Topic models such as LDA take a different view of what a group is (Blei, Ng, and Jordan, 2003). Instead of placing each paper in one group, they describe every paper as belonging partly to several topics at once, for example half to one topic and half to another. That description may well be the more truthful one, but a paper can only stand in one room, so the assignment must be whole or nothing.

Table 6.2: Common alternatives and the required property each one lacks.

Method	What this task needs that it does not give
K-means	a fixed result: it starts from random positions
HDBSCAN	a group for every paper: it declares some papers noise
LDA	one group per paper: it gives each paper several topics

6.3 How Many Groups

The tree is cut into one group per ten papers, which for this conference gives $5,286 / 10$, rounded to 529 groups. This is deliberately far more groups than there are rooms, and the reason lies in the next chapter. If the papers were cut into exactly twelve groups, one per room, the layout would already be decided before the optimization ever ran, and a single oversized group could exceed every room. Cutting into many small groups instead hands Chapter 7 a large set of small, coherent building blocks and the freedom to combine them well.

The choice of ten papers per group balances two failure directions. Toward fewer, larger groups, the blocks become rigid: each one commits many papers at once, the optimizer has fewer ways to fill the rooms tightly, and the largest blocks risk not fitting anywhere. Toward more, smaller groups, the clustering stops doing its job: in the extreme of one paper per group, the placement problem is back to its original 5,286 pieces and nothing was gained. Groups of around ten sit between the two: the clustering does the cheap and safe part of the work, joining papers that clearly belong together, and leaves the difficult judgement of what shares a room to the optimizer.

Because the count is computed from the number of papers rather than chosen by hand, the same rule carries over to any conference the tool is given. The implementation also

computes silhouette scores for cuts between 5 and 50 groups as background; the decision was not based on them, for the reason given in the next section.

6.4 The Result

Table 6.3 summarizes the outcome of cutting the tree. The 529 groups average ten papers each, exactly as intended, the smallest holds 2 papers and the largest 44, and no paper is left alone in a group of one. Every paper sits in exactly one group, so the first two requirements are met by construction. Each group is then represented by its centre, the average of its papers’ vectors, and these centres are the objects the next chapter works with: two groups whose centres lie close are two groups that would sit well in the same room.

Table 6.3: The 529 groups in numbers.

Papers	5,286
Groups	529
Smallest group	2 papers
Largest group	44 papers
Average group	10.0 papers
Groups of one paper	0
Mean cohesion	0.796
Cohesion range	0.692 – 0.925
Groups with cohesion ≥ 0.75	484 of 529
Silhouette (3,000-paper sample)	0.045

The quality of the groups is measured by their cohesion: for each group, the average closeness of its papers to the group’s centre, where 1 would mean the papers all point the same way. The mean cohesion across the 529 groups is 0.796, the loosest group still reaches 0.692, and 484 of the 529 groups sit at 0.75 or higher, so the groups are genuinely about one thing each. This can also be checked by eye. Every group is labelled with the most typical words of its abstracts, and Table 6.4 shows a sample: the largest group is recognisably the literature on bandit algorithms, the next two are graph neural networks and reinforcement-learning-based reasoning, and even the loosest group of all still has a readable theme.

One number in Table 6.3 reads poorly and is reported anyway: the silhouette score of 0.045. The silhouette rewards two things at once: papers being close within their group, and the groups being far apart from each other (Rousseeuw, 1987). The groups here are not far apart, and that is by design: cutting 5,286 papers into 529 groups splits one broad topic into several small neighbouring groups, and neighbours sit close, so a measure that rewards distance between groups must score this low.

Table 6.4: Example groups: the three largest, the tightest, and the loosest.

Group	Size	Cohesion	Most typical words
495	44	0.805	regret, bandit, bandits, bound
188	29	0.771	graph, gnns, node, graphs
426	27	0.820	reasoning, rl, rlvr, models
161	2	0.912	carbon, ai, operational, footprint
475	12	0.692	social, users, collective, agent

Chapter 7

Room Assignment

7.1 The Ideal: A Proven Best Layout

With the papers clustered, one thing is left: deciding which groups share which room. The ideal way to make such a decision is full optimization, meaning a method that searches the problem exactly and returns two things at once. It is the best possible layout, and a mathematical proof that no better layout exists. The proof is what makes this the gold standard. Without a proof, every layout leaves the same doubt: maybe more searching would have found a better one, and nobody can say how much better. With a proof, the doubt is gone: the search stops because nothing better exists. Exact methods for this class of problem exist (Grötschel and Wakabayashi, 1989), and modern solvers are regularly applied to it (Du et al., 2022), so trying the exact way first is the normal choice. This chapter starts with that attempt, then states the problem, reports what happened, and explains the road taken instead.

7.2 The Problem, Written Out

The notation of Chapter 3 carries over. There are $C = 529$ groups with sizes $s_1, \dots, s_C$, and $R = 12$ rooms with seat capacities $c_1, \dots, c_R$. The table W holds the similarity between every pair of group centres, computed from the embeddings of Chapter 5. A layout assigns every group i to exactly one room, written $a_i \in \{1, \dots, R\}$, and every room must hold at least m papers, with $m = 1$ here, so that no room stands empty.

The goal is a single number to be made as large as possible: the sum of the similarities of every pair of groups that share a room. Two similar groups placed together add a large amount to this sum; two unrelated groups placed together add almost nothing. Maximizing the sum therefore pushes similar groups into the same rooms, which is exactly what a topically

coherent session means. Formally,

$$f(a) = \sum_{\substack{1 \leq i < j \leq C \\ a_i = a_j}} W_{ij}, \quad (7.1)$$

and for the exact solver the same problem is written as an integer program, using $y_{ir} = 1$ if group i is in room r , and $z_{ijr} = 1$ if groups i and j share room r :

$$\begin{aligned} \max \quad & \sum_{i < j} \sum_r W_{ij} z_{ijr} \\ \text{s.t.} \quad & \sum_r y_{ir} = 1 \quad \forall i, \quad \sum_i s_i y_{ir} \leq c_r \quad \forall r, \quad \sum_i s_i y_{ir} \geq m \quad \forall r, \\ & z_{ijr} \leq y_{ir}, \quad z_{ijr} \leq y_{jr}, \quad y_{ir}, z_{ijr} \in \{0, 1\}. \end{aligned}$$

Table 7.1 states what every line means in plain words, so that the model can be read without the symbols.

Table 7.1: The optimization model, line by line.

Piece	What it says
$f(a) = \sum W_{ij}$ over shared rooms	the score: total similarity of all group pairs sharing a room
$\max \sum_{i < j} \sum_r W_{ij} z_{ijr}$	make that score as large as possible
$\sum_r y_{ir} = 1$	every group sits in exactly one room
$\sum_i s_i y_{ir} \leq c_r$	no room takes more papers than it has seats
$\sum_i s_i y_{ir} \geq m$	no room is left empty
$z_{ijr} \leq y_{ir}, z_{ijr} \leq y_{jr}$	a pair only counts as together when both groups are in that room
$y, z \in \{0, 1\}$	every decision is a plain yes or no

Two facts about this model matter for everything that follows. First, the problem is a known hard one: it belongs to the family of clique partitioning problems, which are NP-hard (Grötschel and Wakabayashi, 1989), meaning no method is known that solves large cases quickly. Second, the case here is genuinely large. With 529 groups there are 139,656 pairs, and with twelve rooms the model contains over 1.6 million yes-or-no decisions. The number of possible layouts is astronomically beyond what could ever be listed and checked.

7.3 The Exact Attempt

The exact model was implemented in CP-SAT, the constraint solver of Google’s OR-Tools library (Perron and Didier, 2025), and run on the full problem. After four hours it had still

not finished. The run was stopped without a proven result, and nothing suggested it would finish in any acceptable time. No exact experiment was conducted beyond this attempt. This outcome matches what the literature reports: exact solvers struggle on problems of this type as they grow (Du et al., 2022). The attempt settled the direction of the project, though not because anything is known about what organisers prefer. No survey was conducted on whether they would wait days for a proven best layout or trade some quality for speed, and this work does not claim to know. What is known is the nature of the job itself: the details of a conference change while it is being organised — rooms are added or lost, papers are withdrawn, capacities shift — and every such change makes the previous layout outdated. A tool useful under these conditions must be cheap to run again, so that the organiser can experiment, try a change, see its effect, and try another. A method that needs hours for a single answer cannot support this way of working, whatever the quality of that answer. What the task needs is a very good layout in minutes, produced as often as the details change. That is a different requirement, and it is met by a different family of methods.

7.4 The Turn to Heuristics

A heuristic is a method that searches for good solutions without promising the best one. It trades away the proof: instead of certainty, it offers a strong layout within whatever time it is given, and in practice this trade is the standard one for large hard problems (Blum and Roli, 2003). The trade fits this project for two reasons. First, the proof adds little here. The organiser’s question is “give me a very good layout now”, not “prove that nothing better exists”. Conference papers also tend to share close topics, so many groups can stand in for one another and many different layouts reach almost the same quality; when very good layouts are this abundant, the step from very good to proven best changes very little. The tool of Chapter 8 must be cheap to run. A layout that costs a few minutes on ordinary hardware could realistically be offered as a cloud service without large compute expenses; one that costs hours of solver time could not. Third, these methods have a strong record on this problem class (Palubeckis, Ostreika, and Tomkevičius, 2014), so choosing them is established practice, not a gamble.

Heuristics come in distinctly different families, each built around a different core idea (Blum and Roli, 2003): some restart many times from fresh starting points, some take one long walk guided by memory, some escalate how hard they shake a stuck solution, some breed a population of solutions, and some hand parts of the problem back to an exact solver. Five methods were chosen for the comparison, one established representative of each family: GRASP for restarting (Feo and Resende, 1995), Tabu Search for the memory-guided walk (Glover and Laguna, 1997), Variable Neighborhood Search for escalating shakes (Mladenović and Hansen, 1997), a Memetic algorithm for the population approach (Moscato and Cotta, 2003), and a Matheuristic for the hybrid with the exact solver (Maniezzo, Stützle, and Voß, 2010). Covering the families rather than many variants of one idea means the conclusion

of this chapter is about kinds of search, not about one lucky implementation, and each representative is the traditional, widely cited method of its family.

The comparison itself was built to be fair, and the rules are simple. Every method attacked the same problem: the same 529 groups, the same twelve rooms, the same score $f(a)$, the same rule that no room may be empty. Every method received the same time: 60 seconds of search per attempt, a length found to work well for experimentation after trying several values. Every method made twenty attempts, and the attempts were numbered with the same twenty seed values across all methods, so that attempt number three of one method can be paired against attempt number three of another. This pairing is what makes the statistical tests of Section 7.6 valid. One property of these methods makes the fixed time essential, unlike the exact solver, which stops when it has its proof, a heuristic has no natural finishing point and would search forever, so the stopping time is a decision made from outside.

7.5 The Five Methods

Each method is now described in turn: its idea in plain words, its settings, and its results over the twenty attempts. In every results table, the similarity is the average closeness of the papers that ended up sharing a room, and its random-assignment baseline on this problem is 0.304, so every method's layout should be read against that floor.

7.5.1 GRASP

GRASP builds a layout quickly, polishes it, throws it away, and starts again (Feo and Resende, 1995). The building step places groups one by one, and at each placement it chooses randomly among the few best-looking options rather than always the single best, so every restart produces a different starting layout. The polishing step then improves the layout by moving single groups between rooms and swapping pairs of groups until no such move helps any more. The best layout seen across all restarts is kept. Within the 60-second budget, the method completed 40.5 restarts per attempt on average.

Table 7.2: GRASP over 20 attempts of 60 seconds.

Best score	8,306.15
Average score	8,176.36
Spread (std)	76.6
Paper similarity	0.3431
Restarts per attempt	40.5

7.5.2 Tabu Search

Tabu Search improves one layout in a single long walk (Glover and Laguna, 1997). At every step it tries all small changes, moving one group to another room or swapping two groups, and takes the best of them. The catch is that a walk taking only improving steps soon reaches a dead end where no change helps. Tabu Search therefore also accepts worsening steps to walk out of dead ends, and to stop itself from simply undoing them again and circling, it forbids each move for the 12 steps after it is made. The best layout seen anywhere along the walk is kept.

Table 7.3: Tabu Search over 20 attempts of 60 seconds.

Best score	8,279.70
Average score	7,595.77
Spread (std)	333.3
Paper similarity	0.3486
Memory length (tenure)	12

7.5.3 Matheuristic

The Matheuristic mixes the two worlds: a heuristic that steers, and the exact solver doing the detailed work (Maniezzo, Stützle, and Voß, 2010). It rests on one fact from Section 7.3: CP-SAT fails on all 529 groups, yet solves twelve groups perfectly in a fraction of a second. So the method feeds it twelve at a time. It freezes the whole layout, frees twelve groups, and lets CP-SAT place those twelve as well as they can possibly be placed. It locks that answer in, frees a different twelve, and solves again. 754 of these little perfect solves per attempt on average. The bet: many small perfect decisions add up to one strong layout.

Table 7.4: Matheuristic over 20 attempts of 60 seconds.

Best score	8,326.37
Average score	7,893.15
Spread (std)	302.6
Paper similarity	0.3374
Exact sub-solves per attempt	754

7.5.4 Variable Neighborhood Search

Variable Neighborhood Search is built on one idea: when stuck, shake harder (Mladenović and Hansen, 1997). A shake means moving a few randomly chosen groups to random rooms,

deliberately and without checking whether it helps. The method polishes a layout with the same local moves as the other methods until nothing helps, then shakes it and polishes again. If the shake did not lead anywhere better, the next shake is larger, up to ten groups at once, dropping back to a single group whenever an improvement is found. Each attempt completed 121.3 of these shake-and-polish rounds on average.

Table 7.5: VNS over 20 attempts of 60 seconds.

Best score	8,286.03
Average score	7,610.40
Spread (std)	329.6
Paper similarity	0.3493
Shake rounds per attempt	121.3

7.5.5 Memetic Algorithm

The Memetic algorithm keeps a population of 20 layouts at once, each one a complete candidate seating plan (Moscato and Cotta, 2003). It improves them by breeding. Two layouts are picked as parents, and a child is built from them: for each group, the child takes the room that one parent or the other gave it, so the child is a new layout mixing both. Small random changes, called mutations, are then applied to the child, and, the memetic signature, the child is polished with local search before it may join the population. The weakest layouts are dropped, the next generation begins, and each attempt ran 119.5 generations on average. The best layout ever present in the population is kept.

Table 7.6: Memetic algorithm over 20 attempts of 60 seconds.

Best score	8,312.05
Average score	8,150.35
Spread (std)	110.2
Paper similarity	0.3432
Population size / generations	20 / 119.5

7.6 The Choice: GRASP

Table 7.7 compares the five methods under the same conditions. No proven optimum exists, so the gaps are measured against the best layout any method found, the Matheuristic's 8,326.37.

Table 7.7: All five methods over the same 20 attempts of 60 seconds. Gaps are to the best layout found, not to a proven optimum.

Method	Best	Average	Spread	Gap best	Gap avg.	Similarity
GRASP	8,306.15	8,176.36	76.6	0.24%	1.80%	0.3431
Tabu Search	8,279.70	7,595.77	333.3	0.56%	8.78%	0.3486
Matheuristic	8,326.37	7,893.15	302.6	0.00%	5.20%	0.3374
VNS	8,286.03	7,610.40	329.6	0.48%	8.60%	0.3493
Memetic	8,312.05	8,150.35	110.2	0.17%	2.11%	0.3432

All five methods reach roughly the same peak: their best attempts lie within about half a percent of each other. They differ in what they deliver on an ordinary attempt. GRASP and Memetic average 8,176 and 8,150 with small spreads, while Tabu Search and VNS average far lower with spreads about four times larger. One pattern runs the other way and is examined in Chapter 9: the two weakest methods on the objective produce the highest paper similarity.

Whether these differences are real was tested with the standard procedure for repeated algorithm comparisons (Demšar, 2006). Friedman’s test confirms that the methods are not all equal ($\chi^2 = [\text{TO FILL}]$, $p = [\text{TO FILL}]$), and pairwise Wilcoxon tests with Holm correction (Holm, 1979; García and Herrera, 2008) show which pairs differ, in Table 7.8. Nine of the ten pairs differ significantly. The exception is GRASP against Memetic, at $p = 0.355$: the two leaders cannot be separated from each other, and both are better than the remaining three.

Table 7.8: Pairwise Wilcoxon tests, Holm-corrected. Values below 0.05 mean a real difference.

Pair		p (Holm)
GRASP	Memetic	0.355
GRASP	Matheuristic	0.003
GRASP	Tabu Search	< 0.001
GRASP	VNS	< 0.001
Memetic	Matheuristic	0.007
Memetic	Tabu Search	< 0.001
Memetic	VNS	< 0.001
Matheuristic	Tabu Search	< 0.001
Matheuristic	VNS	< 0.001
Tabu Search	VNS	0.043

The choice between them therefore rests on everything else, and all of it favours GRASP: the higher average of the two, 8,176 against 8,150, the smaller spread, 76.6 against 110.2, and a far simpler design, since GRASP builds, improves and restarts where Memetic needs a population, selection, breeding, mutation and local improvement. Where results cannot

separate two methods, the simpler one is the safer choice. GRASP is therefore the method the final tool uses, and how it is configured there is described in Chapter 8.

Chapter 8

The Layout Tool

8.1 The Tool as a Whole

Every chapter up to this point settled one decision on its own. This chapter is where all of them come together into the program that was actually built and handed over.

What the tool does is simple to state. It takes a file of papers and a list of room sizes, and it gives back a seating plan. That is all the user has to provide, and none of the pipeline has to be understood to use it. A whole run is one command:

```
python main.py --papers my-papers.csv --rooms 700,600,550
```

If the program is started without those details, for example by pressing the run button of an editor, it asks for them instead of stopping with an error. The papers file can also be dragged straight into the terminal window, which types its path automatically. Windows and Mac paste such a path in two different ways, and both are cleaned up so that either one works.

What happens after that is one run of six steps, and those six steps are this whole dissertation carried out in order. First, the input is checked and the tool reports what it is about to do. Then every abstract is turned into a vector, using the model chosen in Chapter 5. After that the vectors are grouped: the tree of Chapter 6 is built and cut into small groups of similar papers. Next those groups are packed into the rooms with the method chosen in Chapter 7, which is run several times over so that the best result can be kept. The finished layout is then measured against a random assignment of the same shape. The last step is saving, where three files are written out: the seating plan itself, a summary of every room, and a short quality report. Nothing in between needs a human decision. Table 8.1 shows the whole sequence.

Table 8.1: One run of the tool from start to finish, and where each step was decided.

Step	Goes in	Comes out	Decided in
Check	papers file, room sizes	a usable input	this chapter
Embed	5,286 abstracts	5,286 vectors	Chapter 5
Cluster	5,286 vectors	529 groups	Chapter 6
Pack	529 groups, 12 rooms	a layout	Chapter 7
Measure	the layout	similarity and baseline	Chapter 3
Save	all of the above	three output files	this chapter

Two things are true of the tool on every run. The first is that it sets itself up. The two numbers that shape a run, how many groups to make and how long to search for, are worked out from the size of the file it was given rather than asked of the user, so the same program handles a small workshop and a large conference without being changed.

The second is that it repeats itself. This matters for two reasons. A tool that gives a different answer every time cannot really be checked, since an organiser who reruns it and gets something else has no way of telling whether the change came from something they did or from the tool itself. And every number reported in this dissertation has to be reproducible by anyone who runs the same files again. Table 8.2 goes through the run and separates the parts that come out the same every time from the one part that does not.

Table 8.2: What is fixed in a run of the tool, and what is not.

Part of the run	Why
<i>Fixed, the same on every run</i>	
Checking the input	rows are dropped by rule, not by chance
Working out the settings	both numbers come from the paper count
Embedding	the same abstract always gives the same vector
Clustering	Ward has no random step in it
Which layouts get built	the random choices are set in advance
Measuring the layout	plain arithmetic on the finished layout
The random baseline	the shuffles are seeded and then averaged
<i>Not fixed</i>	
How many get built	the search stops on the clock, so a faster computer gets through more of them

Only one part of the run is not fixed, and it is worth being clear about what that does and does not mean. The random choices inside the search are set in advance, so the layouts always come out in the same order: the first one built is always the same layout, the second one is always the same, and so on down the list. What is not fixed is how far down that list the tool gets, because it stops when the time is up rather than after a set number of tries. A

faster computer works through more layouts in the same 53 seconds than a slower one does. If the best layout of the run happens to be far down the list, a fast machine reaches it and a slow machine does not.

In practice this has not caused a difference. The same conference was run three times and all three runs gave back the same plan, with a paper similarity of 0.3395 against a baseline of 0.3040 each time. A stopping rule based on a fixed number of tries rather than on the clock would remove the last piece of variation altogether, at the cost of a run whose length can no longer be predicted, and this is picked up in Chapter 10.

The program is split across five files, listed in Table 8.3. Only `main.py` knows anything about files, folders and users. The other four are handed data and give data back, and they never open a file themselves. The reason for splitting it this way is that each step can then be replaced on its own: a different embedding model, or a different optimisation method, means changing one file, and nothing else in the tool needs to know about it.

Table 8.3: The five files of the tool and what each one does.

File	What it does
<code>main.py</code>	reads the input, checks it, runs the four steps, writes the results
<code>embeddings_miniLM.py</code>	turns every abstract into a vector
<code>hierarchical_clustering.py</code>	builds the tree and cuts it into groups
<code>GRASP.py</code>	packs the groups into the rooms
<code>layout_evaluation.py</code>	measures the finished layout

8.2 Input

The tool is meant for conference organisers rather than programmers. Implementation was completed with the purpose of making it as seamless as possible. Before any of the heavy process starts it checks whether the given input is correct for the task. Anything it finds wrong is explained in plain words, not left as a crash for the user to decode.

The first check is the papers csv file. It has to exist, and it has to have a column of abstracts. Table 8.4 lists the four columns the tool looks for, and the abstract is the only one it cannot manage without. A file holding nothing but a column of abstracts runs perfectly well; the other three only make the finished plan easier to read.

Table 8.4: The columns the tool looks for in the papers file.

Column	Needed?	What it is for
Abstract	yes	the text everything is computed from
Title	no	names the papers in the seating plan
Category	no	labels the rooms in the summary
ID	no	numbers the papers; row numbers are used otherwise

Finding the abstract column is made flexible, since paper files rarely name it the same way twice. It is called **Abstract** by default, and the name is matched while ignoring capitals and any spaces around it, so **abstract** and **ABSTRACT** are both picked up without the user having to say anything. If the user's csv abstract column is named something else entirely, the user can inform the program of the new naming convention. Adding `--abstract-col` followed by the right naming convention tells the program what to look for, so for example, a file whose abstracts sit in a column called **Summary** is run like this:

```
python main.py --papers my_papers.csv --rooms 700,600,550 --abstract-col "Summary"
```

The tool then looks for **Summary** instead of **Abstract**, and the program works with the same instructed way. The same process of adding new naming convention can also be done to the other columns as follows:

```
python main.py --papers my_papers.csv --rooms 700,600,550
--abstract-col "Summary" --category-col "Field" --id-col "PaperNo"
```

Three things can go wrong with the abstract column, and the tool answers each in proportion to how bad it is. If no matching column is found, it cannot guess which one was meant, so it prints every column name the file does have and stops; seeing the real names is usually enough to correct the command. If the column is there but some of its cells are empty, those papers are dropped, since a blank abstract cannot be turned into a vector, and the tool prints how many it dropped rather than removing them quietly. If every cell is empty, nothing is left to work with and the run stops. The rule behind all three is the same: the tool stops only when it truly cannot continue, and whenever it can carry on it says exactly what it did.

The next checks are about the rooms. Room sizes have to be positive whole numbers, so a typo like 0 or a negative number is caught here. Then come the two checks that matter most, both about whether the papers and the rooms actually fit together. The rooms have to have at least as many seats as there are papers, since every paper needs somewhere to go, and if there are not enough the tool prints how many papers and how many seats there are and stops. The number of papers also has to be at least the number of rooms, since every room has to end up with at least one paper. These two catch opposite problems: the first stops a run with too many papers for the space, the second a run with too few papers to

fill the rooms, which is what would happen if eight papers were handed twelve rooms. Only once all of this passes does the tool proceed with its functions.

Two more columns are sought for, but only to make the output readable. A column named `Title` is used to name the papers in the seating plan. A column of categories is looked for under the names conferences usually give it, such as *primary area*, *category*, *track* or *topic*, and it can also be named directly with a flag if it is called something unusual. The reason it is worth collecting is that it gives the organiser a quick way of judging the layout by eye. The room summary then shows the three most common categories in every room, so a glance is enough to see whether each room came out as one recognisable subject or as a mixture. These labels are never read by the tool itself, which means that the result whether good or bad, is a result produced by the tool simply based on abstract data.

8.3 Hierarchical Clustering

Two numbers shape a run, and the tool works both of them out from the file it was given. Nothing is asked of the user.

The first number is how many groups to make. The rule comes from Chapter 6: one group per ten papers. The tool adds two safety lines around it. It never makes fewer groups than there are rooms, since every room needs at least one group. And it never makes more groups than there are papers.

The second number is how long each try may search. The rule is one second per hundred papers, so more papers means more time. Two cut-offs sit around it. A very small file still gets at least 10 seconds, because anything shorter is too little search to produce a decent layout. A very large file gets at most 300 seconds per try, because anything longer would leave the user waiting on a run for half an hour or more. In between, the rule runs as it is: this conference has 5,286 papers, so every try gets 53 seconds. One thing to note is why the time is set differently here than in Chapter 7. The experimentation needed fairness. Five methods were being compared, so every method had to get exactly the same time, and it was fixed by hand at 60 seconds. The tool needs something else. It has to work on any conference it is given, so the time has to come from the input itself. The rule is one second per hundred papers, which puts this conference at 53 seconds, close to the 60 that worked well in the tests. Any other size gets its time in the same proportion, meaning twice the papers, twice the time. Table 8.5 shows what both rules come to on the two datasets the tool has been run on.

Table 8.5: The rules the tool works out by itself, and what they come to on each dataset.

Rule	NeurIPS (5,286 papers)	arXiv sample (10,000 papers)
groups = papers / 10	529 groups	1,000 groups
time = papers / 100 seconds	53 s per try	100 s per try
tries, best one kept	5	5
minimum papers per room	1	1

Chapter 6 already showed how the papers are grouped, and the tool changes nothing about that. The tree is built the same way and cut by the same rule of one group per ten papers. So the job of this step inside the tool is different. Grouping well was settled there. Here the step prepares everything the packing will touch.

The packing that follows never sees a paper. It only sees groups, and each group is shrunk down to two things, a size and a direction. The size says how many seats the group needs, and every room is checked against it so nothing overflows. The direction is one single vector for the whole group, made by averaging the vectors of its papers, and it stands for what the group is about. This shrinking is the reason of why our program spends less computational resources in the optimization step. The packing handles a few hundred of these instead of thousands of papers, and that is why it can finish in seconds.

Before the next pipeline step starts, the clustering phase also prepares the answers for the heuristics part. It will keep asking one question, how alike are these two groups. So the answer is worked out once for every pair of groups, 139,656 pairs here, and written into a table. Two small rules apply to it. Anything below zero is stored as zero, and a group gets zero with itself. Only real togetherness should earn points.

One more thing is remembered from the cut. Every paper's group number is kept, because the packing answers in groups while the final plan must answer in papers. When the papers are split into groups, the tool writes down each paper's group number, one line per paper. This list exists because of a gap. GRASP only says where a group goes. The final file must say where a paper goes. The list closes the gap. GRASP puts group 17 in Room 3, the list shows every paper with the number 17, and each of those papers is written into the file with Room 3 beside it. And because the list points at the paper's own row in the original table, its title and category are picked up from there and written into the same line. Done for every group, the result is the seating plan, one row per paper with its title, its room, and its field.

A small repair rounds the step off. A group larger than the largest room would make the whole task impossible, since it fits nowhere. If the cut ever produces one, the same tree is cut a little finer and checked again, until nothing oversized remains. Cutting a finished tree again is nearly free. On the two datasets of this work this repair never had to run.

8.4 Heuristics

The packing is done by GRASP, which was chosen in Chapter 7. This is how it runs inside the tool.

GRASP works by making a complete layout, then throwing it away and making another one. Each of these attempts is called a try. A try starts from nothing, places every group into a room, improves what it has made, and ends with one finished layout that could be handed to an organiser as it stands. The tool then keeps whichever try came out best. The reason for working this way rather than carefully building one layout is that a single attempt can easily settle into a poor arrangement, and starting fresh costs so little that it is worth doing many times over.

One try is made of two parts: building a layout, then improving it. The building part places whole groups into rooms, one group at a time, and it works through them from the biggest group down to the smallest. Big groups are the hardest to fit anywhere, so they get their pick while the rooms are still empty. It runs in two stages.

Stage one fills every room once. It goes through the rooms in turn and, for each one, puts in the biggest group that still fits. This is what keeps any room from ending up empty.

Stage two places everything that is left. For each group, the tool looks at every room that still has space and scores it: the score is how similar this group is to the groups already sitting in that room. A room already holding related work scores high, a room holding unrelated work scores low. The group then goes into one of the three highest-scoring rooms, picked at random rather than always the top one.

That random pick has a specific purpose. Without it, every try would follow the same choices and build exactly the same layout, so running five tries would give five identical answers and there would be nothing to choose between. Picking among the top three instead means each try takes a slightly different path and produces a different layout. Occasionally a try runs itself into a dead end, where a group is left over and no room has space for it. When that happens the half-built layout is thrown away and a new one is started.

The improving part takes the finished layout and makes small changes to it for as long as they help. There are two kinds of change: moving one group into a different room, and swapping two groups between their rooms. A change is kept only if it raises the score, and only if it leaves every room legal, meaning inside its capacity and not below its minimum. When neither kind of change helps any more, the layout is done.

Building and improving then start over inside the same try, again and again, until the time for that try is up. Each of these repeats is a restart, and the best layout any of them produced is what the try returns. Chapter 7 measured about forty restarts in a minute of search, so a try is not one layout but the best of a few dozen.

The tool runs five of these tries and keeps the best of the five. Five is enough because a user needs one good layout, not a fair comparison of methods, and a few long tries serve that better than many short ones. Running five also guards against one unlucky try, since

GRASP is random and any single try can come out lower than the rest. Each try starts from its own fixed number, so running the tool twice on the same file gives back the same five layouts.

One thing to note for reading the results in Chapter 9. GRASP is handed a function that measures paper similarity and it reports that number for every try, but it never optimises on it. What GRASP maximises is the group-level score from Chapter 7, and the paper similarity is only carried along for the report.

8.5 What the Tool Gives Back

Every run makes its own numbered folder, **Production 1**, **Production 2** and so on, and the tool counts upwards until it finds a number that is not taken yet. What this does is make sure no run can ever overwrite an earlier one, so a user who tries three different sets of room sizes still has all three results afterwards. Three files are written into the folder.

The first is `layout.csv`, which is the actual deliverable: one row per paper, giving its number, its title, the room it goes to, and its category if the file had one. The second is `room_summary.csv`, one row per room, giving its capacity, how many papers it got, how full it is, how many groups it holds, the average similarity of the papers inside it, and the three most common categories in that room. The third is `quality_report.txt`, a short summary of the whole run, which for the NeurIPS conference looks like this:

```
source: neurips_accepted_submissions.csv
papers: 5286 rooms: 12 seats: 5286
clusters: 529 (~10 papers each)
method: grasp, 5 tries x 53s, best kept
average within-room paper similarity: 0.3395
random-assignment baseline: 0.3040
lift over random: +0.0355
```

The purpose of the report is mostly on its last three lines. A similarity number alone says nothing, because nobody knows what counts as high. So the tool also shuffles the papers into the same rooms at random, measures that, and prints both numbers with the gap between them. The gap shows the actual success and quality of our method. In one of the datasets, the whole tool run itself took five tries of 53 seconds, a little over four minutes of searching, plus a few seconds for embedding and clustering.

That result can also be held against the benchmark of Chapter 7, where GRASP was measured over twenty tries of 60 seconds and averaged a paper similarity of 0.3431. The tool reaches 0.3395 while spending about a fifth of that search time, so the settings it uses cost a small amount of quality and buy a lot of speed. The rule behind them is a starting point that was calibrated rather than proven: it was set so that a conference of this size gets a

budget close to the one that worked in the benchmark, and what it gives to conferences of very different sizes has not been measured. This is picked up again in Chapter 10.

The tool searches for less time than the experiment did and gets almost the same quality. The benchmark on Chapter 7 averaged 0.3431 over twenty tries of 60 seconds. The tool reaches 0.3395 with about a fifth of that search time. A fifth of the time, 99 percent of the result. A good trade.

The rule deciding the search time, one second per hundred papers, was tuned on this one conference. For any other size the rule guesses by proportion, and those guesses have not been checked. Chapter 10 returns to this, and to the fix of letting the user choose the search time themselves.

8.6 The Demonstration

The tool was run on two datasets. The first one is the conference this whole work was built on. The second one is the demonstration set from Section 4.4, ten thousand papers taken from nine different fields of arXiv. Nothing in this dissertation was decided based on this second set. Both files were handed over exactly as they came, with no cleaning done to them. What this section does is show what came out, read straight from the tool’s own output files. What these results mean is left for Chapter 9.

Table 8.6 starts by putting the two runs next to each other, using the headline numbers from each run’s quality report.

Table 8.6: The two runs of the tool.		
	NeurIPS	arXiv sample
Papers	5,286	10,000
Rooms	12	12
Seats	5,286	10,800
Groups made	529	1,000
Search per try	53 s	100 s
Similarity	0.3395	0.1735
Random baseline	0.3040	0.0658
Lift over random	+0.0355	+0.1077

Each run also writes a file called room summary, and the two tables below print those files as they are, with the original column names and the original values. What each column holds is simple. `capacity` is the seats of the room. `fill_pct` says how full it came out. `n_clusters` says how many groups it holds. `avg_paper_similarity` is the average closeness of the paper pairs inside it. `top_categories` gives the three subject labels that appear most among its papers, together with their counts. It stays in the files. Another thing to note is that both input files

carried a subject label for every paper, the NeurIPS primary areas and the arXiv categories, and the tool found both columns on its own.

Table 8.7 is the file from the NeurIPS run. The input was `neurips_accepted_submissions.csv` with 5,286 papers, twelve rooms holding 5,286 seats, 529 groups, and GRASP with five tries of 53 seconds. Since the seats add up to exactly the number of papers, there was no spare room anywhere, and every room came out completely full.

Table 8.7: `room_summary.csv` as produced for the NeurIPS run.

room	capacity	papers	fill_pct	n.clusters	avg_paper_similarity	top_categories
Room 1	700	700	100.0	135	0.2655	deep_learning (162) applications (120) theory (117) applications (235)
Room 2	600	600	100.0	68	0.3663	deep_learning (182) machine_learning_for_sciences (68) deep_learning (186)
Room 3	550	550	100.0	59	0.3503	social_and_economic_aspects_of_machine_learning (77) applications (71) theory (146)
Room 4	500	500	100.0	64	0.3113	optimization (75) deep_learning (62) applications (199)
Room 5	450	450	100.0	34	0.4286	deep_learning (166) general_machine_learning (23) deep_learning (114)
Room 6	425	425	100.0	39	0.3036	theory (59) probabilistic_methods (55) deep_learning (190)
Room 7	400	400	100.0	33	0.3664	machine_learning_for_sciences (50) social_and_economic_aspects_of_machine_learning (45) deep_learning (232)
Room 8	380	380	100.0	24	0.4296	applications (63) social_and_economic_aspects_of_machine_learning (26) applications (145)
Room 9	350	350	100.0	18	0.3540	deep_learning (114) social_and_economic_aspects_of_machine_learning (33) deep_learning (115)
Room 10	320	320	100.0	15	0.3469	social_and_economic_aspects_of_machine_learning (61) theory (33) deep_learning (112)
Room 11	300	300	100.0	18	0.3379	probabilistic_methods (40) applications (39) reinforcement_learning (73)
Room 12	311	311	100.0	22	0.3318	deep_learning (50) optimization (43)

Table 8.8 is the file from the demonstration run. The input was `papers_dataset_random_10K.csv` with 10,000 papers, twelve rooms holding 10,800 seats, 1,000 groups, and GRASP with five tries of 100 seconds. This time the rooms hold 800 more seats than there are papers. Nine rooms still came out completely full, Room 12 came out 91 percent full, and Room 9 ended up holding a single group of four papers.

Table 8.8: room_summary.csv as produced for the demonstration run.

room	capacity	papers	fill_pct	n_clusters	avg_paper_similarity	top_categories
Room 1	1300	1300	100.0	175	0.1586	stat.CO (162) stat.ME (152) econ.EM (147)
Room 2	1200	1200	100.0	100	0.1754	eess.IV (256) eess.AS (218) stat.ML (99)
Room 3	1100	1100	100.0	101	0.2413	q-fin.PR (123) q-fin.ST (102) q-fin.RM (102)
Room 4	1000	1000	100.0	78	0.1757	econ.TH (224) econ.GN (182) econ.EM (110)
Room 5	950	950	100.0	121	0.1625	nlin.SI (191) nlin.CD (82) nlin.PS (70)
Room 6	900	900	100.0	87	0.2006	q-bio.MN (108) q-bio.PE (83) q-bio.CB (82)
Room 7	850	849	99.9	68	0.2010	math.GT (45) math.AG (42) math.AC (42)
Room 8	800	800	100.0	80	0.1719	nlin.PS (134) nlin.AO (89) nlin.CD (75)
Room 9	750	4	0.5	1	0.3801	econ.GN (4) eess.SY (193)
Room 10	700	700	100.0	75	0.1299	eess.SP (102) cs.OS (30) eess.SP (47)
Room 11	650	650	100.0	82	0.0801	astro-ph.EP (28) q-bio.TO (27) eess.AS (66)
Room 12	600	547	91.2	32	0.0664	nlin.CD (47) stat.OT (23)

One thing matters when reading the last column. The pipeline never read any of these labels. It only read the abstracts. So when a whole room comes out as one field, the abstracts alone put it there. In the demonstration run this happened again and again, finance in Room 3, economics in Room 4, biology in Room 6, mathematics in Room 7. In the NeurIPS run the rooms mix areas instead, and deep learning shows up in almost every room. Why the two runs look this different, why the spare seats gathered in Room 9 and Room 12, and how the numbers of the two runs compare, is looked at in Chapter 9.

Chapter 9

Results, Discussion and Analysis

9.1 The Final Layout

9.2 Per-Room Analysis

9.3 The Cost of Exact Fill

9.4 The Ranking of Methods Depends on the Budget

9.5 Objective and Similarity Do Not Always Agree

9.6 Interpreting the Magnitude

9.7 Against Expectations and Prior Work

Chapter 10

Conclusion

10.1 Summary

10.2 Limitations

10.3 Future Work

Appendix A

Supplementary Material

A.1 Room Capacities

A.2 Full Benchmark Tables

A.3 Cluster Examples

A.4 Code and Reproducibility

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